

Unit-4

System testing: -

- System testing is level of testing
- System Testing is a type of testing technique in which the entire system is tested.
- Software testing is a process of executing a program with the primary goal of finding errors. If we want to eliminate errors from our software, we need to conduct testing.
- In this testing, the behavior of the entire software product is checked based on the Software Requirement Specification (SRS).
- The main objective of this testing is to evaluate the needs of the business and end-users.
- This is a type of black box testing, where the external structure of the software is tested. In this, it is not necessary for the tester to have knowledge of the internal structure of the software.
- System testing categories based on who is doing the testing?

The objectives of system testing are:

- System testing is performed as one of the initial levels of testing in the Software Development Life Cycle (SDLC). Here, the entire system is thoroughly tested.
- To systematically uncover different types of errors in minimum time and with a minimum amount of effort.
- To ensure that the system is reliable and working as per the specifications.
- To ensure that the system is according to the requirements of the users.
- To ensure that the system accepts correct input and gives out correct and accurate output. The system should be able to detect and avoid any incorrect input, output and processing.
- To ensure high quality performance level

Test Plan

- A test plan is a detailed document which describes software testing areas and activities.
- It outlines the test strategy, objectives, test schedule, required resources (human resources, software, and hardware), test estimation and test deliverables.
- The test plan is a base of every software's testing.
- A test plan is a document that describe the scope, approach, resources, and schedule of testing activities.

Importance of Test Plan:

- **Understanding Testing Details:** The Test Plan document helps stakeholders outside the test team, such as developers, business managers, and customers, understand the details of testing.
- **Acting as a Rule Book:** The Test Plan serves as a rule book that guides the testing process. It provides a structured approach to follow during testing activities.
- **Documentation of Important Aspects:** The Test Plan documents important aspects such as test estimation, test scope, and test strategy. These documented details can be utilized in other projects as well.
- **Defining Roles and Responsibilities:** Test Plan document describe the roles and responsibilities of each team member.

Types of Test Plan: -

There are three types of the test plan:

- ✓ Master Test Plan.
- ✓ Phase Test Plan.
- ✓ Testing Type Specific Test Plans.

Master Test Plan:

- This type of test plan performs multiple levels of testing like unit testing integration testing, system testing and integrates the entire testing strategy.

Phase Test Plan:

- This test plan outlines the testing strategy for a specific phase of the project. For example, it may include a list of tools, list of test cases, etc.

Testing Type Specific Test Plans:

- This test plan is designed for a specific type of testing, such as security testing, load testing, performance testing, etc. In other words, a specific test plan is designed for non-functional testing.

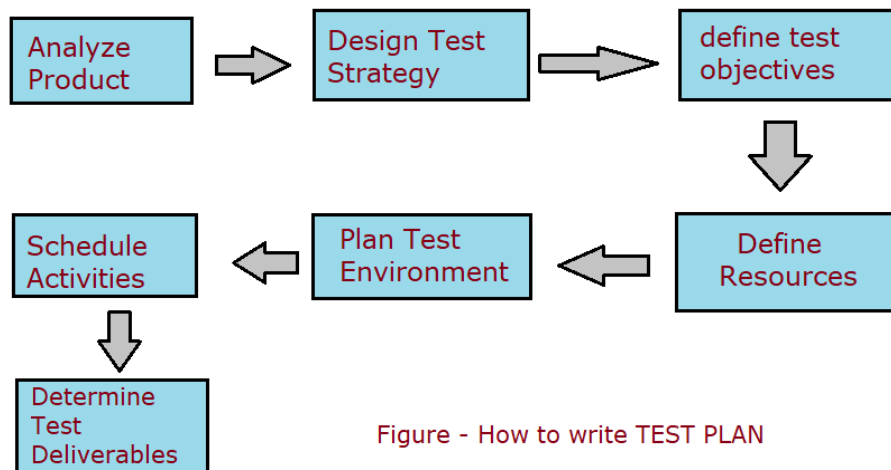


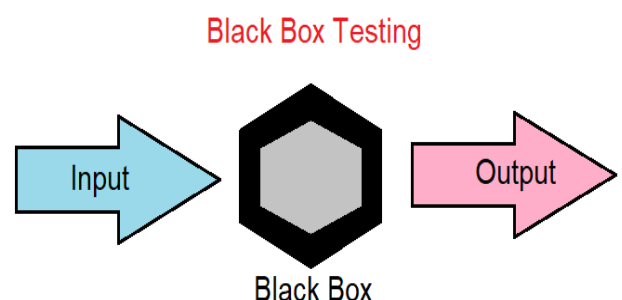
Figure - How to write TEST PLAN

Types of testing:

There are two categories of testing: -

1. Black box testing

- **Black box testing** is a software testing technique in which the functionality of the software is tested, and the tester does not need to have knowledge of the software's internal structure.
- In other words, "Black box testing is a testing method where the tester does not require knowledge of the internal structure of the software, and coding knowledge is also not necessary."
- Black box testing is also referred to as **input-output testing**. It is called black box testing because the software under test is like a black box to the tester, meaning the tester cannot see inside it.
- Testers do not need to know the internal workings of the software.



2. White box testing

- The testing in which the tester has knowledge of the software's internal structure is called white box testing. This type of testing is based on the internal workings of the software

- White box testing is named as such because the software is like a white box to the tester, meaning the tester can see inside it. In other words, "White box testing is a software testing technique in which the internal structure, design, and coding of the software are tested."
- In this type of testing, the tester has knowledge of programming.
- Since the tester can see the code in this testing, it is also referred to as clear **box testing**, **transparent box testing**, and **glass box testing**.

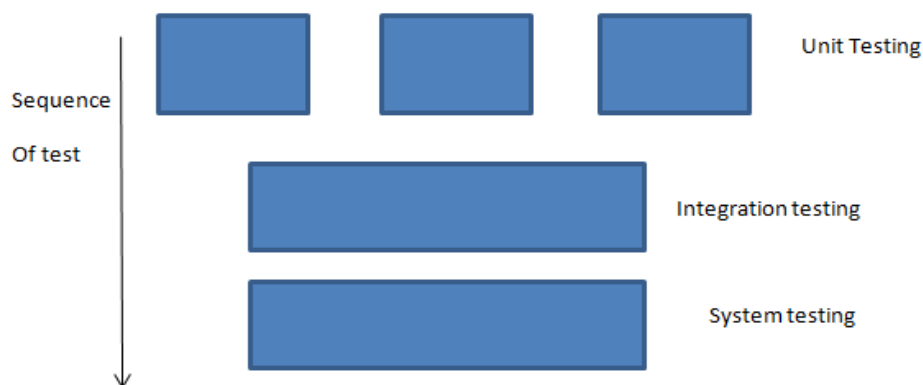
Various/types levels of testing

Testing is a set of activities. These activities must be planned and conducted systematically so that it must not leave any scope for rework.

In order to conduct a proper testing, the following order should be performed:

- Unit testing
- Integration testing.
- System testing.

Unit testing focuses on the testing of individual sub component. After unit testing, sub component is integrated in component. Errors often arise on integration. After integration testing, system tests are performed that focus on the overall system



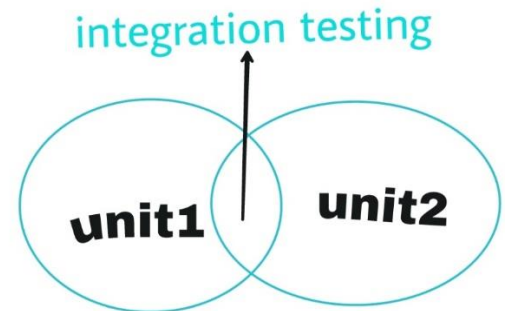
Unit Testing: -

- Unit Testing is a type of functional testing where each unit or component of software is tested individually.

- It is performed throughout the coding phase of software development and is primarily done by programmers.
- It is also referred to as component testing.
- Unit Testing is a methodology of software testing where the smallest parts of a software application (referred to as units) are tested.

Integration testing: -

- In this Testing, all units and components are grouped together and tested as a whole. The main objective is to ensure that these units interact correctly with each other.
- Integration testing is a systematic technique that integrates the tested sub component, find errors, remove them and check the program structure as specified by design.

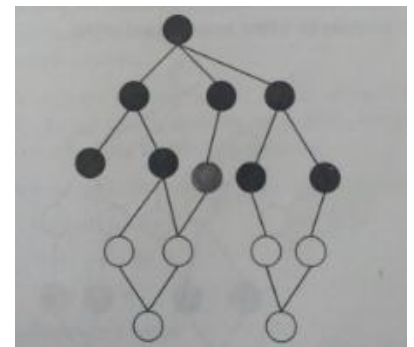


Two basic types of integration are usually used:

- Top-down integration
- Bottom up integration

(i) Top-down Integration

- Top-down integration is an incremental approach that requires the highest-level modules be tested and integrated first.
- Add one or more other modules. Continue the process until all modules have been integrated and tested.



Advantages of Top-down Integration

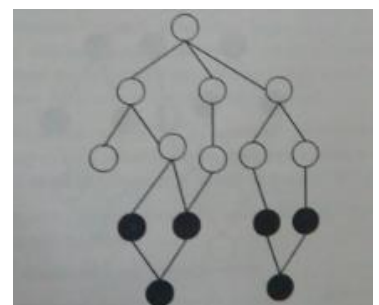
- Top-level interfaces are tested first.
- Top-level routines provide a natural test harness for lower-level routines.

Disadvantages of Top-down Integration

- Delays verification of low-level modules.
- Provides poor support for early release of limited functionality.

(ii) Bottom up Integration

The bottom up integration approach requires the lowest level modules be tested and integrated first. Add one or more additional modules. Continue the process until all modules have been integrated and tested.



Advantages of Bottom up Integration

- Developing and testing starts with the actual modules.
- Allows early verification of low-level modules.

Disadvantages of Bottom up Integration

- Provides poor support for early release of limited functionality.
- Delays verification of high level behavior.

3. System Testing

- System testing implies testing of the entire software. The goal of system testing is to see if the system meets its requirements.
- System testing includes a series of different tests to fully exercise the system.
- Each test has a different purpose and the series of tests verify that all system elements are properly integrated and are performing the allocated functions.

System testing involves the following types of testing:

- (i) Recovery testing
- (ii) Security testing
- (iii) Stress testing

Recovery Testing

- Recovery testing is designed to examine how easily and completely the system can recover a system failure.
- Recovery testing forces the system to fail in a variety of ways and verifies that system recovery is properly performed. A system with quick recovering capability and with minimum human intervention is desirable.

(ii) Security testing

Security testing tests or verifies the:

- Protection mechanism built into a system to determine that the system is protected from unauthorized personnel.
- Other systems cannot gain access to the system and information within it.

(iii) Stress Testing

- Stress Testing is a testing technique that verifies the stability and reliability of software.

- Its main objective is to determine when the software will crash under heavy load.
- Additionally, it helps determine the software's capacity to handle loads. Heavy load refers to tasks such as storing a large amount of data or opening numerous programs simultaneously.
- This testing also known as **load testing**.

System Implementation: -

- System implementation is the process of turning plans and designs into a working system or software solution.
- It involves installing, configuring, testing, and deploying the system to make it operational.
- It involves translating the design and architecture of the system into a working, functional product that meets the specified requirements
- system implementation is a complex process that requires careful planning, coordination, and execution to ensure the successful deployment of a new system or software solution.

Importance of System implementation

- Ensures that the system meets the needs and requirements of users.
- Helps organizations achieve their goals and objectives.
- Improves efficiency and productivity by streamlining processes.
- Maximizes the value of investments in new technologies.

Implementation process-

1.Planning and Preparation:

- Before implementation begins, thorough planning and preparation are essential. This includes finalizing requirements, allocating resources, creating a detailed implementation plan, and establishing communication channels among stakeholders. In planning we will include:
 - Define project scope, objectives, and requirements.
 - Develop an implementation plan with tasks and timelines.
 - Identify stakeholders and establish communication channels.
 - Assess potential risks and develop mitigation strategies

2.Installation:

- The first step in implementation is installing the necessary hardware and software components.
- This may involve setting up servers, networks, databases, and other infrastructure required for the system to operate.

3.Configuration:

- Once the hardware and software are installed, they need to be configured according to the system design and requirements.
- This involves setting parameters, adjusting settings, and integrating different components to ensure they work together seamlessly.

4.Data Migration:

- If the new system is replacing an existing one, data migration is often necessary.
- This involves transferring data from the old system to the new one while ensuring data integrity and consistency.
- Data cleansing and validation may also be performed during this stage.

4.Testing:

- Once the system is installed and configured, thorough testing is conducted to ensure that it functions as intended and meets the specified requirements.
- This includes functional testing, performance testing, security testing, and user acceptance testing.

5.Training:

- Proper training is essential to ensure that users understand how to use the new system effectively.
- Training sessions may be conducted for end-users, administrators, and other stakeholders to familiarize them with the system's features and functionality.

6.Deployment:

- After testing and training are complete, the system is ready for deployment.

- Depending on the organization's policies and procedures, deployment may be done all at once (big bang) or in stages (phased rollout) to minimize disruption to operations.

7. Monitoring and Maintenance: Once the system is live, ongoing monitoring and maintenance are necessary to ensure its continued performance and reliability. This includes monitoring system metrics, addressing any issues or bugs that arise, and applying updates and patches as needed.

Types of Implementation

There are three types of implementation:

1. Fresh Implementation
2. Replacement Implementation
3. Modified Implementation

Fresh Implementation:

- This refers to the deployment of a completely new system or software where there was no previous version in place.
- It involves setting up the system from scratch and configuring it according to the requirements.

Replacement Implementation:

- In this type of implementation, a new system or software is deployed to replace an existing one.
- The old system may be outdated, inefficient, or unable to meet current needs, so it is replaced with a newer version or a completely different solution.

Modified Implementation:

- Modified implementation involves making changes or modifications to an existing system or software.
- This could include adding new features, updating existing functionality, or customizing the software to better suit the organization's evolving needs.